

B3 Quadratic Functions

Name: _____

Class: _____

Date: _____

Time: **350 minutes**

Marks: **295 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Show that the equation

$$10 \operatorname{cosec}^2 x = 16 - 11 \cot x$$

can be written in the form

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

(1)

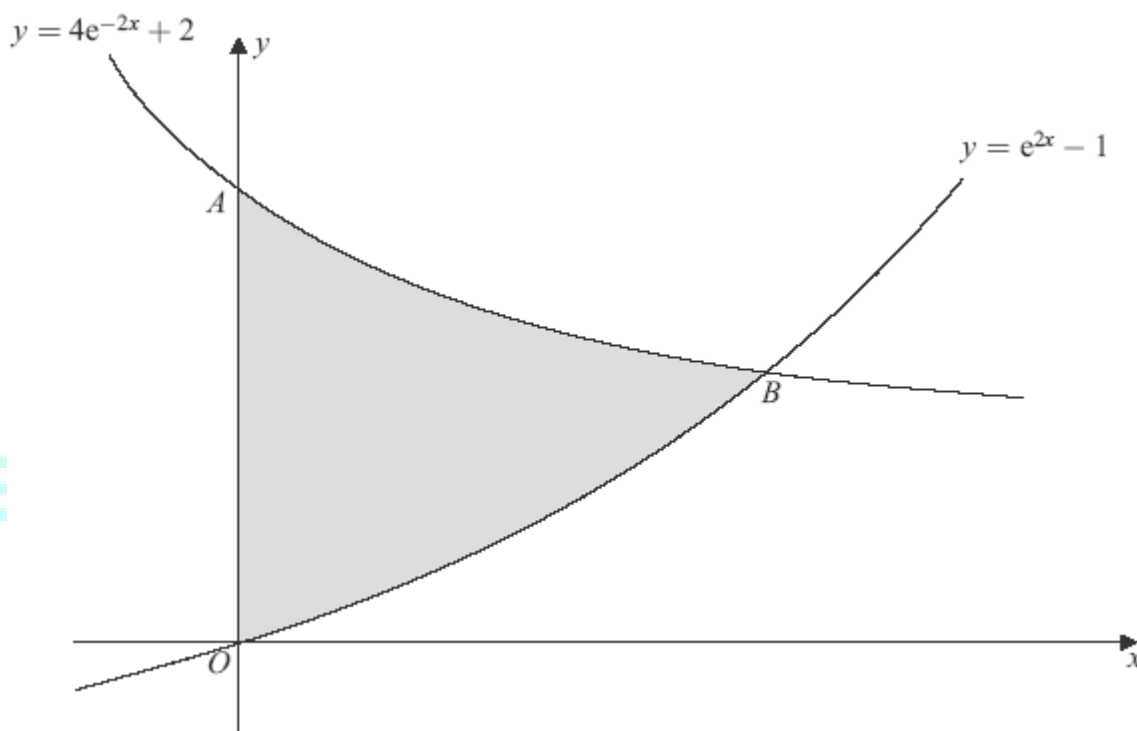
- (b) Hence, given that $10 \operatorname{cosec}^2 x = 16 - 11 \cot x$, find the possible values of $\tan x$.

(4)

(Total 5 marks)

Q2.

The diagram shows the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$.



The curve $y = 4e^{-2x} + 2$ crosses the y -axis at the point A and the curves intersect at the point B .

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} - 1$.

(4)

- (b) Write down the coordinates of the point A .

(1)

- (c) (i) Show that the x -coordinate of the point B satisfies the equation

$$(e^{2x})^2 - 3e^{2x} - 4 = 0 \quad (2)$$

- (ii) Hence find the exact value of the x -coordinate of the point B . (3)

- (d) Find the exact value of the area of the shaded region bounded by the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$ and the y -axis.

(5)
(Total 15 marks)

Q3.

- (a) (i) Express $x^2 + 2x + 5$ in the form $(x + p)^2 + q$, where p and q are integers. (2)

- (ii) Hence show that $x^2 + 2x + 5$ is always positive. (1)

- (b) A curve has equation $y = x^2 + 2x + 5$.

- (i) Write down the coordinates of the minimum point of the curve. (2)

- (ii) Sketch the curve, showing the value of the intercept on the y -axis. (2)

- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 2x + 5$. (3)

(Total 10 marks)

Q4.

- (a) Given that $3e^x = 4$, find the exact value of x . (2)

- (b) (i) By substituting $y = e^x$, show that the equation $3e^x + 20e^{-x} = 19$ can be written as $3y^2 - 19y + 20 = 0$. (1)

- (ii) Hence solve the equation $3e^x + 20e^{-x} = 19$, giving your answers as exact values. (3)

(Total 6 marks)

Q5.

- (a) (i) Express $x^2 - 8x + 17$ in the form $(x - p)^2 + q$, where p and q are integers. (2)
- (ii) Hence write down the minimum value of $x^2 - 8x + 17$. (1)
- (iii) State the value of x for which the minimum value of $x^2 - 8x + 17$ occurs. (1)
- (b) The point A has coordinates $(5, 4)$ and the point B has coordinates $(x, 7 - x)$.
- (i) Expand $(x - 5)^2$. (1)
- (ii) Show that $AB^2 = 2(x^2 - 8x + 17)$. (3)
- (iii) Use your results from part (a) to find the minimum value of the distance AB as x varies. (2)
- (Total 10 marks)**

Q6.

The curve C has equation $y = k(x^2 + 3)$, where k is a constant.

The line L has equation $y = 2x + 2$.

- (a) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation

$$kx^2 - 2x + 3k - 2 = 0$$

(1)

- (b) The curve C and the line L intersect in two distinct points.

- (i) Show that

$$3k^2 - 2k - 1 < 0$$

(4)

- (ii) Hence find the possible values of k .

(4)

(Total 9 marks)

Q7.

- (a) Given that $2 \ln x = 5$, find the exact value of x .

(1)

- (b) Solve the equation

$$2 \ln x + \frac{15}{\ln x} = 11$$

giving your answers as exact values of x .

(5)

(Total 6 marks)

Q8.

A circle with centre C has equation $x^2 + y^2 - 10y + 20 = 0$.

- (a) By completing the square, express this equation in the form

$$x^2 + (y - b)^2 = k$$

(2)

- (b) Write down:

- (i) the coordinates of C ;

(1)

- (ii) the radius of the circle, leaving your answer in surd form.

(1)

- (c) A line has equation $y = 2x$.

- (i) Show that the x -coordinate of any point of intersection of the line and the circle satisfies the equation $x^2 - 4x + 4 = 0$.

(2)

- (ii) Hence show that the line is a tangent to the circle and find the coordinates of the point of contact, P .

(3)

- (d) Prove that the point $Q(-1, 4)$ lies inside the circle.

(2)

(Total 11 marks)

Q9.

- (a) Factorise $9 - 8x - x^2$.

(2)

- (b) Show that $25 - (x + 4)^2$ can be written as $9 - 8x - x^2$.

(1)

- (c) A curve has equation $y = 9 - 8x - x^2$.

- (i) Write down the equation of its line of symmetry. (1)
 - (ii) Find the coordinates of its vertex. (2)
 - (iii) Sketch the curve, indicating the values of the intercepts on the x -axis and the y -axis. (3)
- (Total 9 marks)**

Q10.

The curve C has equation $y = x^2 + 7$. The line L has equation $y = k(3x + 1)$, where k is a constant.

- (a) Show that the x -coordinates of any points of intersection of the line L with the curve C satisfy the equation

$$x^2 - 3kx + 7 - k = 0 \quad (1)$$

- (b) The curve C and the line L intersect in two distinct points. Show that

$$9k^2 + 4k - 28 > 0 \quad (3)$$

- (c) Solve the inequality $9k^2 + 4k - 28 > 0$. (4)
- (Total 8 marks)**

Q11.

- (a) Sketch the graph of $y = 3^x$, stating the coordinates of the point where the graph crosses the y -axis. (2)

- (b) Describe a single geometrical transformation that maps the graph of $y = 3^x$:
- (i) onto the graph of $y = 3^{2x}$; (2)
 - (ii) onto the graph of $y = 3^{x+1}$. (2)

- (c) (i) Using the substitution $Y = 3^x$, show that the equation

$$9^x - 3^{x+1} + 2 = 0$$

can be written as

$$(Y - 1)(Y - 2) = 0$$

(2)

- (ii) Hence show that the equation $9^x - 3^{x+1} + 2 = 0$ has a solution $x = 0$ and, by using logarithms, find the other solution, giving your answer to four decimal places.

(4)

(Total 12 marks)

Q12.

The straight line L has equation $y = 3x - 1$ and the curve C has equation

$$y = (x + 3)(x - 1)$$

- (a) Sketch on the same axes the line L and the curve C , showing the values of the intercepts on the x -axis and the y -axis.

(5)

- (b) Show that the x -coordinates of the points of intersection of L and C satisfy the equation $x^2 - x - 2 = 0$.

(2)

- (c) Hence find the coordinates of the points of intersection of L and C .

(4)

(Total 11 marks)

Q13.

- (a) Express $x^2 - 3x + 4$ in the form $(x - p)^2 + q$, where p and q are rational numbers.

(2)

- (b) Hence write down the minimum value of the expression $x^2 - 3x + 4$.

(1)

- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 - 3x + 4$.

(3)

(Total 6 marks)

Q14.

The quadratic equation $(k + 1)x^2 + 4kx + 9 = 0$ has real roots.

- (a) Show that $4k^2 - 9k - 9 \geq 0$.

(3)

- (b) Hence find the possible values of k .

(4)

(Total 7 marks)

Q15.

The line AB has equation $3x + 5y = 8$ and the point A has coordinates $(6, -2)$.

- (a) (i) Find the gradient of AB .

(2)

- (ii) Hence find an equation of the straight line which is perpendicular to AB and which passes through A .

(3)

- (b) The line AB intersects the line with equation $2x + 3y = 3$ at the point B . Find the coordinates of B .

(3)

- (c) The point C has coordinates $(2, k)$ and the distance from A to C is 5. Find the **two** possible values of the constant k .

(3)

(Total 11 marks)

Q16.

The quadratic equation $(k + 1)x^2 + 12x + (k - 4) = 0$ has real roots.

- (a) Show that $k^2 - 3k - 40 \leq 0$.

(3)

- (b) Hence find the possible values of k .

(4)

(Total 7 marks)

Q17.

- (a) Express $\cos 2x$ in terms of $\sin x$.

(1)

- (b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of x .

(2)

- (ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0^\circ < x < 360^\circ$.

(4)

- (c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

(2)

(Total 9 marks)

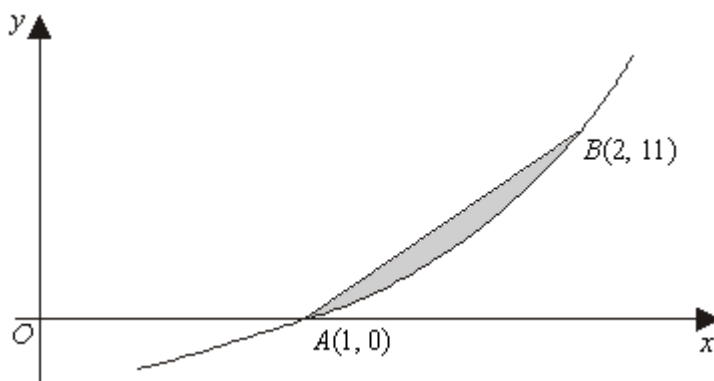
Q18.

- (a) (i) Express $x^2 + 10x + 19$ in the form $(x + p)^2 + q$, where p and q are integers. (2)
- (ii) Write down the coordinates of the vertex (minimum point) of the curve with equation $y = x^2 + 10x + 19$. (2)
- (iii) Write down the equation of the line of symmetry of the curve $y = x^2 + 10x + 19$. (1)
- (iv) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 10x + 19$. (3)
- (b) Determine the coordinates of the points of intersection of the line $y = x + 11$ and the curve $y = x^2 + 10x + 19$. (4)

(Total 12 marks)

Q19.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 + 4x - 5$.
- (i) Use the Factor Theorem to show that $x - 1$ is a factor of $f(x)$. (2)
- (ii) Express $f(x)$ in the form $(x - 1)(x^2 + px + q)$, where p and q are integers. (2)
- (iii) Hence show that the equation $f(x) = 0$ has exactly one real root and state its value. (3)
- (b) The curve with equation $y = x^3 + 4x - 5$ is sketched below.



The curve cuts the x -axis at the point $A(1, 0)$ and the point $B(2, 11)$ lies on the

curve.

(i) Find $\int (x^3 + 4x - 5) \, dx$.

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

(Total 14 marks)

Q20.

The quadratic equation

$$(2k - 3)x^2 + 2x + (k - 1) = 0$$

where k is a constant, has real roots.

(a) Show that $2k^2 - 5k + 2 \leq 0$.

(3)

(b) (i) Factorise $2k^2 - 5k + 2$.

(1)

(ii) Hence, or otherwise, solve the quadratic inequality

$$2k^2 - 5k + 2 \leq 0.$$

(3)

(Total 7 marks)

Q21.

(a) (i) Express $x^2 - 4x + 9$ in the form $(x - p)^2 + q$, where p and q are integers.

(2)

(ii) Hence, or otherwise, state the coordinates of the minimum point of the curve with equation $y = x^2 - 4x + 9$.

(2)

(b) The line L has equation $y + 2x = 12$ and the curve C has equation $y = x^2 - 4x + 9$.

(i) Show that the x -coordinates of the points of intersection of L and C satisfy the equation

$$x^2 - 2x - 3 = 0$$

(1)

(ii) Hence find the coordinates of the points of intersection of L and C .

(4)

(Total 9 marks)

Q22.

The quadratic equation $x^2 + (m + 4)x + (4m + 1) = 0$, where m is a constant, has equal roots.

- (a) Show that $m^2 - 8m + 12 = 0$.

(3)

- (b) Hence find the possible values of m .

(2)

(Total 5 marks)

Q23.

It is given that n satisfies the equation

$$2 \log_a n - \log_a (5n - 24) = \log_a 4$$

- (a) Show that $n^2 - 20n + 96 = 0$.

(3)

- (b) Hence find the possible values of n .

(2)

(Total 5 marks)

Q24.

- (a) Express $x^2 + 8x + 19$ in the form $(x + p)^2 + q$, where p and q are integers.

(2)

- (b) Hence, or otherwise, show that the equation $x^2 + 8x + 19 = 0$ has no real solutions.

(2)

- (c) Sketch the graph of $y = x^2 + 8x + 19$, stating the coordinates of the minimum point and the point where the graph crosses the y -axis.

(3)

- (d) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 8x + 19$.

(3)

(Total 10 marks)

Q25.

A circle has equation $x^2 + y^2 - 4x - 14 = 0$.

- (a) Find:

- (i) the coordinates of the centre of the circle; (3)
- (ii) the radius of the circle in the form $p\sqrt{2}$, where p is an integer. (3)
- (b) A chord of the circle has length 8. Find the perpendicular distance from the centre of the circle to this chord. (3)
- (c) A line has equation $y = 2k - x$, where k is a constant.
- (i) Show that the x -coordinate of any point of intersection of the line and the circle satisfies the equation
- $$x^2 - 2(k + 1)x + 2k^2 - 7 = 0$$
- (3)
- (ii) Find the values of k for which the equation
- $$x^2 - 2(k + 1)x + 2k^2 - 7 = 0$$
- has equal roots. (4)
- (iii) Describe the geometrical relationship between the line and the circle when k takes either of the values found in part (c)(ii). (1)
- (Total 17 marks)

Q26.

- (a) Simplify $(k + 5)^2 - 12k(k + 2)$. (2)
- (b) The quadratic equation $3(k + 2)x^2 + (k + 5)x + k = 0$ has real roots.
- (i) Show that $(k - 1)(11k + 25) \leq 0$. (5)
- (ii) Hence find the possible values of k . (3)
- (Total 10 marks)

Q27.

- (a) Express $x^2 - 6x + 16$ in the form $(x - p)^2 + q$. (2)
- (b) A curve has equation $y = x^2 - 6x + 16$. Using your answer from part (a), or

otherwise:

- (i) find the coordinates of the vertex (minimum point) of the curve; (2)
 - (ii) sketch the curve, indicating the value where the curve crosses the y -axis; (2)
 - (iii) state the equation of the line of symmetry of the curve. (1)
- (c) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 - 6x + 16$. (3)
- (Total 10 marks)**

Q28.

The cubic polynomial $p(x)$ is given by $p(x) = (x - 2)(x^2 + x + 3)$.

- (a) Show that $p(x)$ can be written in the form $x^3 + ax^2 + bx - 6$, where a and b are constants whose values are to be found. (2)
 - (b) Prove that the equation $(x - 2)(x^2 + x + 3) = 0$ has only one real root and state its value. (3)
- (Total 5 marks)**

Q29.

A line has equation $y = mx - 1$, where m is a constant.

A curve has equation $y = x^2 - 5x + 3$.

- (a) Show that the x -coordinate of any point of intersection of the line and the curve satisfies the equation $x^2 - (5 + m)x + 4 = 0$ (1)
- (b) Find the values of m for which the equation $x^2 - (5 + m)x + 4 = 0$ has equal roots. (4)
- (c) Describe geometrically the situation when m takes either of the values found in part (b). (1)

(Total 6 marks)

Q30.

- (a) (i) Show that the equation

$$4 \tan \theta \sin \theta = 15$$

can be written as

$$4 \sin^2 \theta = 15 \cos \theta$$

(1)

- (ii) Use an appropriate identity to show that the equation

$$4 \sin^2 \theta = 15 \cos \theta$$

can be written as

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

(2)

- (b) (i) Solve the equation $4c^2 + 15c - 4 = 0$.

(2)

- (ii) Hence explain why the only value of $\cos \theta$ which satisfies the equation

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

is $\cos \theta = \frac{1}{4}$.

(1)

- (iii) Hence solve the equation $4 \tan \theta \sin \theta = 15$ giving all solutions to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$.

(2)

- (c) **Write down** all the values of x in the interval $0^\circ \leq x \leq 90^\circ$ for which

$$4 \tan 4x \sin 4x = 15$$

giving your answers to the nearest degree.

(2)

(Total 10 marks)

Q31.

(In this question, take $g = 10 \text{ m s}^{-2}$.)

A projectile is fired from a point O with speed u at an angle of elevation α above the horizontal so as to pass through a point P . The projectile travels in a vertical plane through O and P . The point P is at a horizontal distance $2k$ from O and at a vertical distance k above O .

- (a) Show that α satisfies the equation

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0 \quad (7)$$

(b) Deduce that

$$u^4 - 20ku^2 - 400k^2 \geq 0 \quad (3)$$

(Total 10 marks)

Q32.

A ball is projected with speed $u \text{ m s}^{-1}$ at an angle of elevation α above the horizontal so as to hit a point P on a wall. The ball travels in a vertical plane through the point of projection. During the motion, the horizontal and upward vertical displacements of the ball from the point of projection are x metres and y metres respectively.

(a) Show that, during the flight, the equation of the trajectory of the ball is given by

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha) \quad (6)$$

(b) The ball is projected from a point 1 metre vertically below and R metres horizontally from the point P .

(i) By taking $g = 10 \text{ m s}^{-2}$, show that R satisfies the equation

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0 \quad (2)$$

(ii) Hence, given that u and R are constants, show that, for $\tan \alpha$ to have real values, R must satisfy the inequality

$$R^2 \leq \frac{u^2(u^2 - 20)}{100} \quad (2)$$

(iii) Given that $R = 5$, determine the minimum possible speed of projection.

(3)
(Total 13 marks)